Statistics for Social Science

Week 4

The Normal Curve, the Central Limit Theorem, Confidence Intervals and p-Values

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Outline

- Part 1 – A short recap, then the normal curve
  - SD → z-score: where one person sits
  - The normal curve, and why the shape matters
- Part 2 – The Central Limit Theorem
  - Seeing it, and checking it with TGSS age
  - Standard error and the 95% confidence interval
- Part 3 – From the confidence interval to the p-value
  - What p says, and why .05
  - Normality tests in Explore: age and comminute

What You Should Learn Today

1 **What the normal curve is, and what a z-score says about one person**

z = (value − mean) / SD · 0 = the mean · −1.67 = 1.67 SDs below the mean

2 **What the Central Limit Theorem says, in one sentence**

Means of samples form a bell around the true mean – whatever the variable looks like

3 **SD vs SE, and how to read a 95% confidence interval**

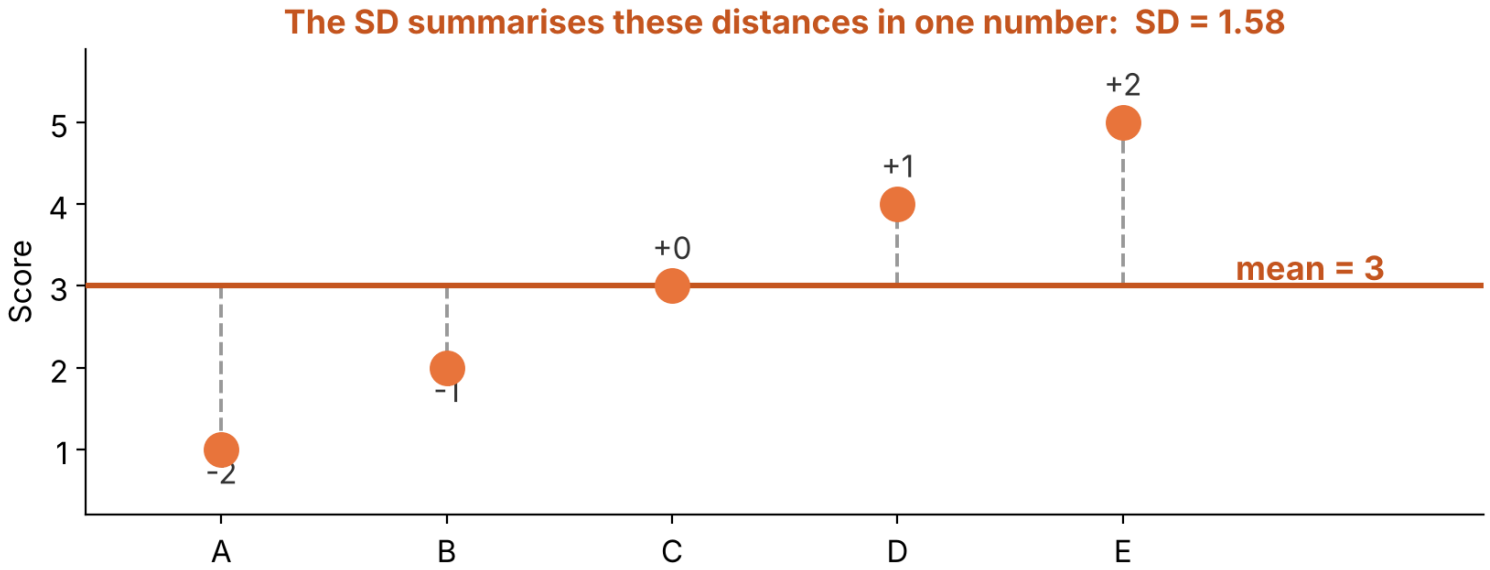
SE = SD ÷ √n · 95% CI = mean ± 1.96 × SE · both are in the Explore output

4 **What a p-value says, and how to test whether a variable is normal**

Small p (≤ .05) = surprising if the idea were true · Shapiro-Wilk and K-S in Explore

Recap 1 – The Standard Deviation

The SD answers one question: **how far is a typical person from the mean?**



$$SD_{sample} = \sqrt{\frac{\sum (x_i - \bar{x})^2}{N-1}}$$

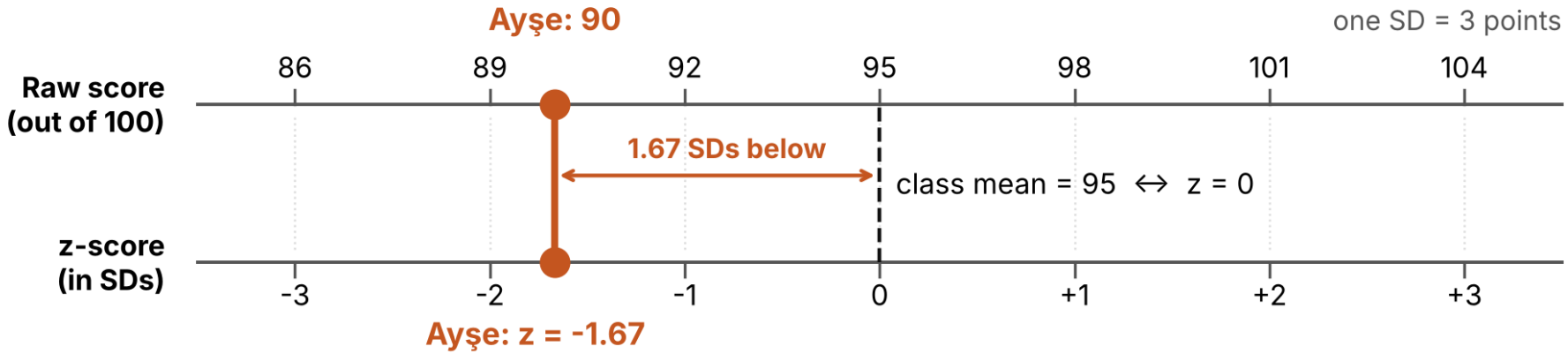
Where:

- SD_{sample} = Sample Standard Deviation
- $\sum$ means "the sum of"
- N = Sample size
- x_i = Each value from the sample
- $\bar{x}$ = The sample Mean

- The smallest possible SD is 0 – everyone has the same score, the group is homogeneous.
- The larger the SD, the more the group differs from itself.
- Same units as the variable; always reported with the mean (M = 167 cm, SD = 10.13 cm).

Recap 2 – From Raw Scores to z

SD: one number for the whole group. **z-score:** one number for each person.



THE POINT

A z-score shows where one person sits in the group

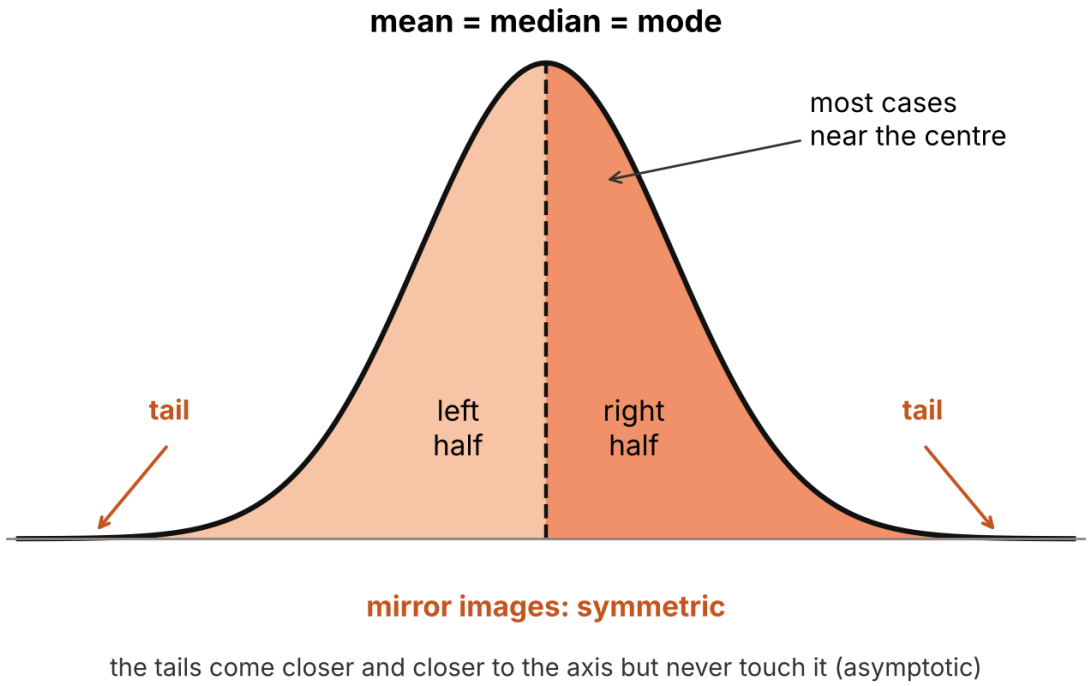
z = (value − mean) / SD the distance from the mean, measured in SDs.

Ayşe: (90 − 95) / 3 = −5 / 3 = −1.67 → she scored 1.67 SDs below the class mean.

Reading z: 0 = at the mean · negative = below · positive = above · size = how many SDs away

The Normal Curve

- A theoretical distribution: a smooth, symmetric, bell-shaped curve
  - Mean = median = mode, all at the centre
  - Symmetric – the left half mirrors the right half
  - Most cases sit close to the centre; very few are far out in the ta
  - The tails never touch the horizontal axis – this property is calle asymptotic
- No real variable is exactly normal. We only ask: is it close enough?



Why Does the Shape Matter?

The shape of your variable decides which analysis you are allowed to use.

If the distribution is…	You describe it with…	And you use…
Approximately normal	Mean and SD	Parametric tests (t-test, ANOVA, Pearson, regression)
Clearly not normal	Median and IQR	Non-parametric tests (Mann–Whitney, Kruskal–Wallis, Spearman)

THE HABIT

Look at the distribution before you choose the analysis

Histogram first, then skewness and kurtosis. This is the step people skip – and it is the step that decides whether the rest of the analysis is defensible.

Population and Sample

There is a **difference** between **population** and **sample.**

Population is everyone we want to say something about – for example, all adults living in Turkey.

Sample is the group we actually observe – for example, the people who answered TGSS 2024.

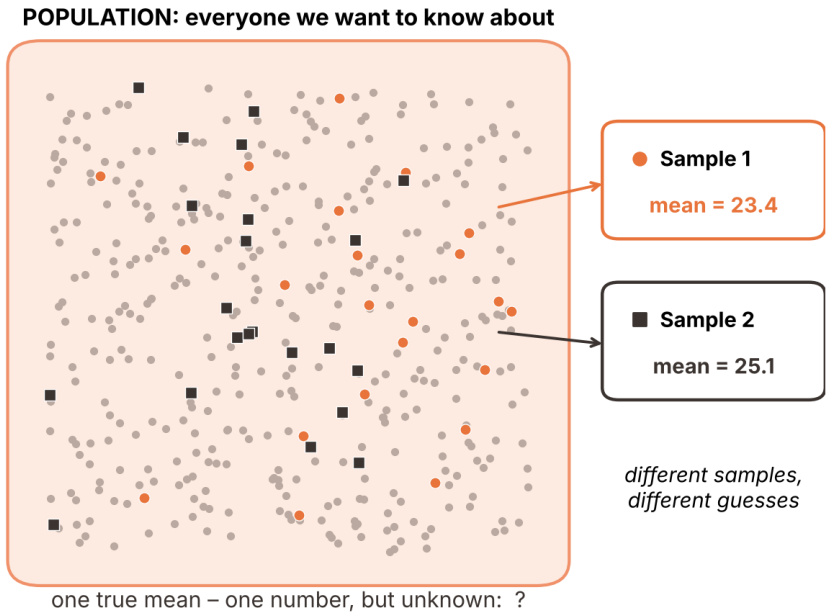
We almost never see the whole population. We see one sample.

- The population has one true mean – but we do not know it.
- Our sample mean is our best guess for it.

Why mean though?

The middle carries the most information about the whole distribution.

In most scenarios, this is the mean.



- Another sample would give a slightly different guess. So instead of one number, we give a range: “the true mean is probably somewhere between __ and __”.
- Our guess misses the true mean by some amount: **the error**. To give a range, we need to know how big this error usually is, and how often it is small or large. That is a question of probability.

Reading Probability from a Shape

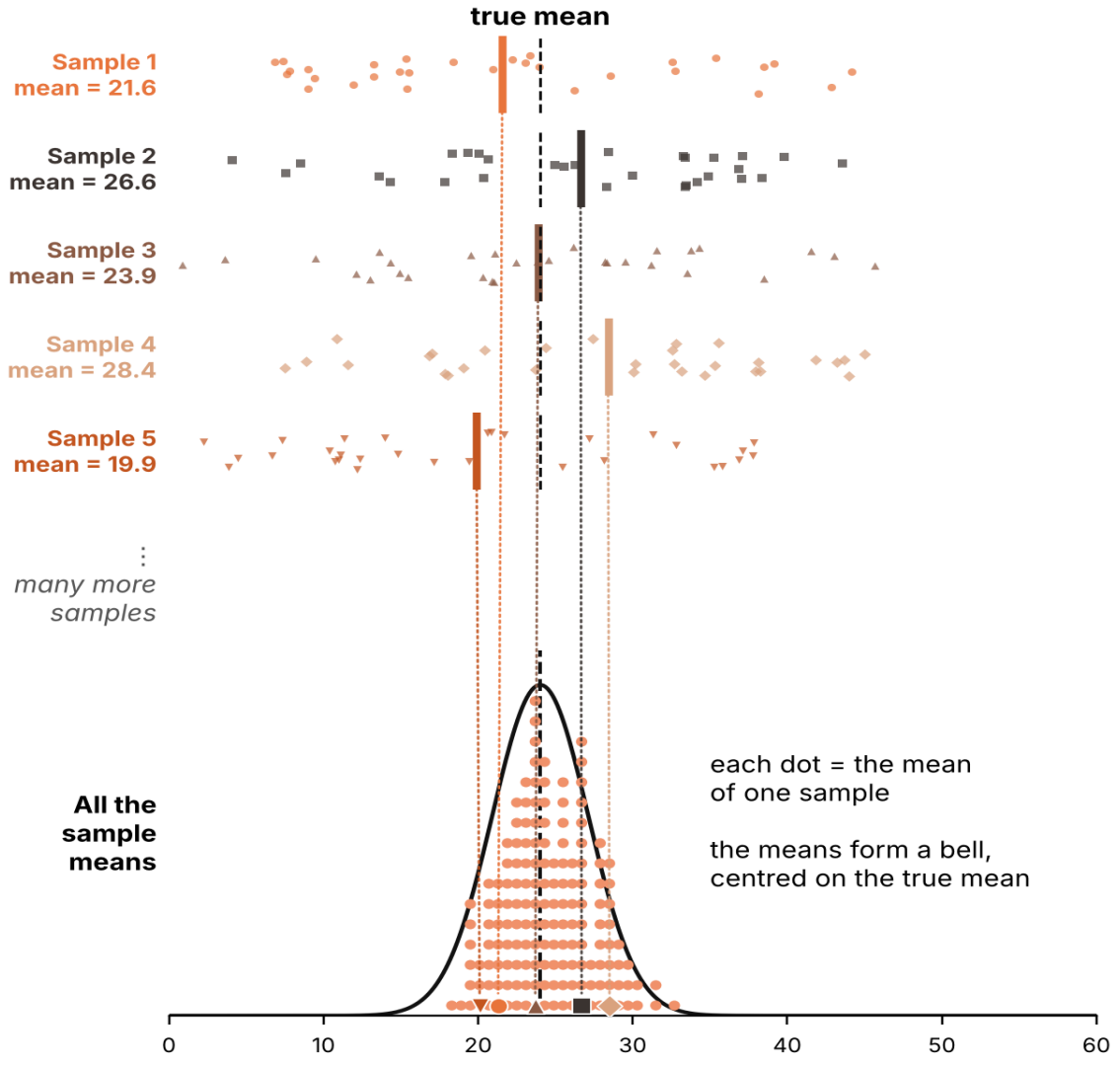
Error = sample mean – true mean.

- We can never calculate our own error, because we do not know the true mean. But we can know the **shape** of all possible errors.
- How? Based on the Central Limit Theorem.

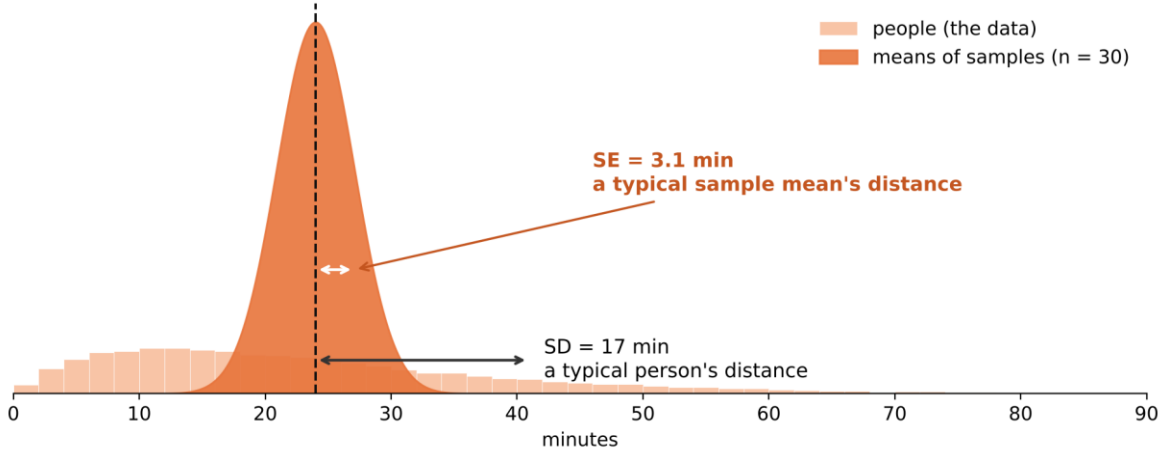
The theorem says: if we take many samples of the same size from a population, and calculate the mean of each sample, then these sample means form a normal distribution.

- This distribution of means is centred on the true population mean.

Let's examine it with an example!



Standard Deviation vs Standard Error



THE FORMULA

SE = SD ÷ √n

SD = 17 min

n = 30 → SE = 17 ÷ 5.5 = 3.1

n = 1700 → SE = 17 ÷ 41.2 = 0.41

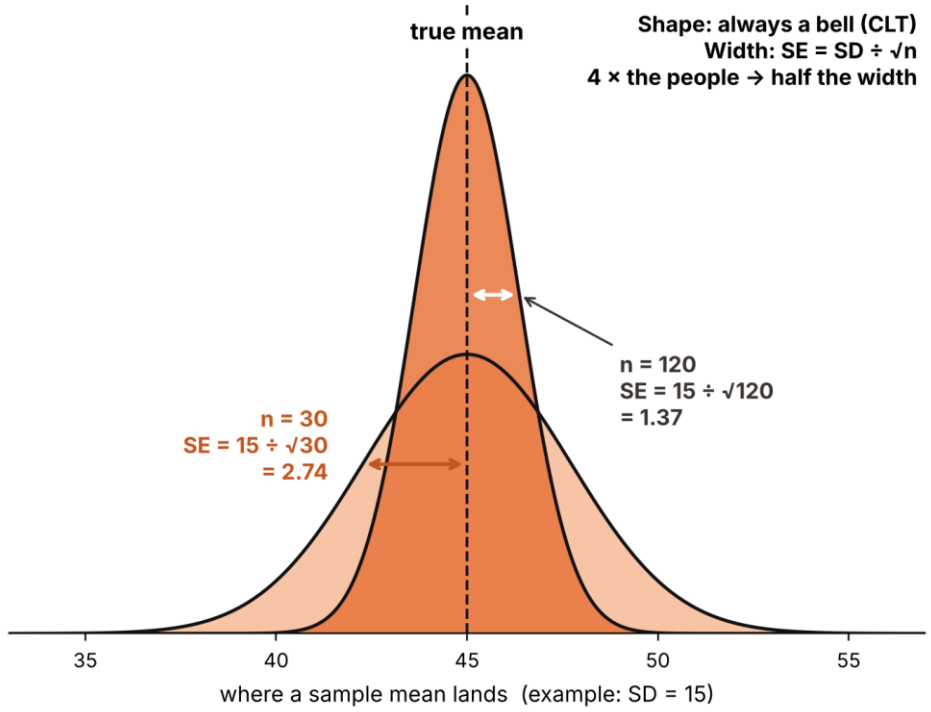
More people, smaller SE: the mean becomes more precise.

	Standard deviation (SD)	Standard error (SE)
Describes	The people in our data	The mean we estimated
Question	How far is a typical person from the mean?	How far is a typical sample mean from the true mean?
Bigger sample?	Stays about the same	Gets smaller
Used for	Describing the data	Inference about the population

One Sample: What Can We Know?

- The CLT tells us the shape: the means of all possible samples always form a bell. So we do not need to draw hundreds of samples.
- What the CLT does not tell us: **how narrow or wide the bell is.**
- We calculate the width from our one sample: **SE = SD ÷ √n**

  SD: how spread out the people in our sample are · n: how many people are in our sample
- The SE tells us how far a sample mean typically lands from the true mean.
- Why √n? Random ups and downs only partly cancel out. To halve the SE, you need four times as many people.
- **Why do we need the SE?** It lets us calculate where the true (population) mean probably lies: the confidence interval.



From SE to a Confidence Interval

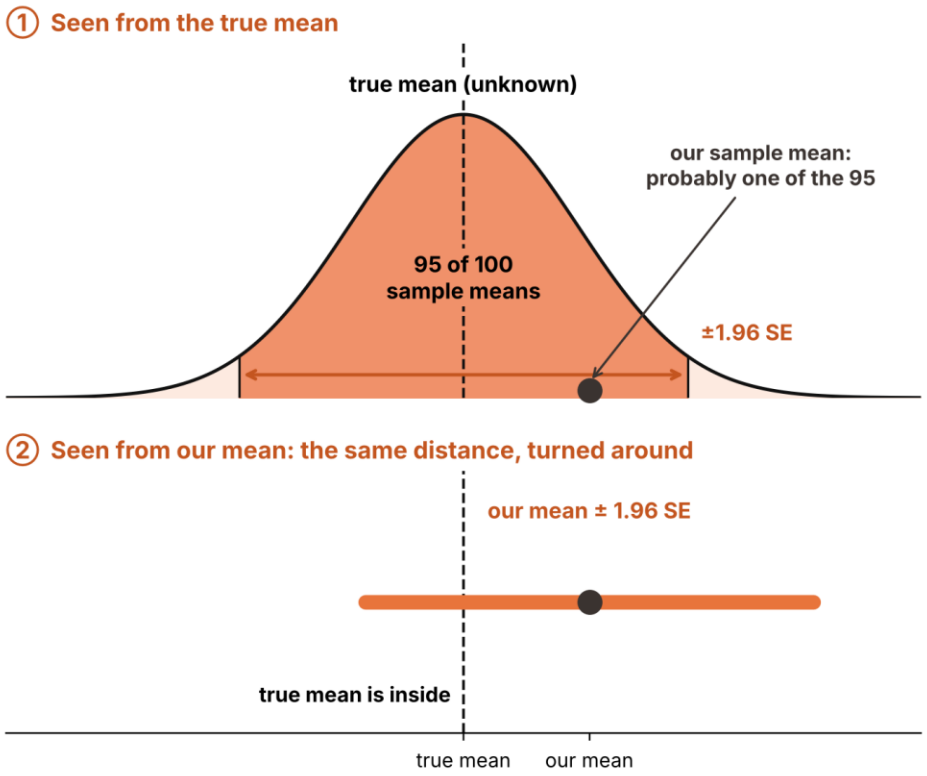
- We know the shape of the errors (a bell, from the CLT) and the typical size of an error (the SE).
- For the 95% range of the true mean: take our mean, subtract 1.96 × SE and add 1.96 × SE. These are the lower and upper bounds.

  95% CI = our mean ± 1.96 × SE

- Example: mean = 45, SE = 2.7 → 1.96 × 2.7 = 5.4 → **39.6 to 50.4**

  "We are 95% confident that the true mean is between 39.6 and 50.4."

- Where does 1.96 come from? In every normal curve, 95% of the values lie between z = −1.96 and z = +1.96. It is a fixed property of the normal curve.
- The 5% outside: 2.5% in each tail. In about 5 of 100 samples the interval misses the true mean. **This 5% is where the .05 of the p-value comes from.**



From the CI to the p-Value

So far we asked: where is the true mean? Now a different question: **someone claims a value – do our data agree?**

Claim: “The average age in Turkey is 39.”

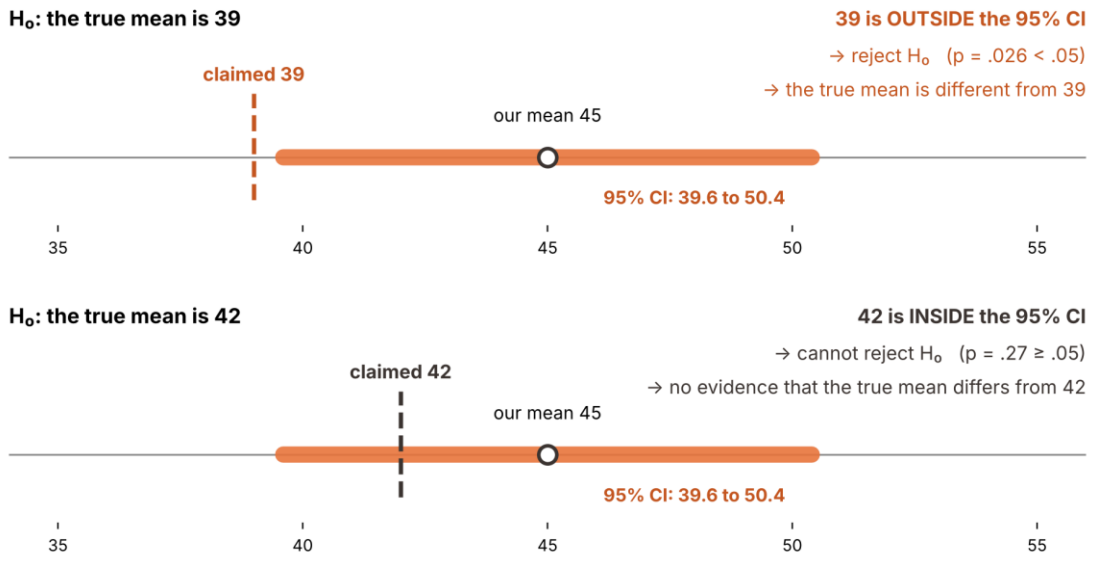
H_0: the true mean is not different from 39. **H_1:** it is different from 39.

We assume H_0 is true and ask: how rare would our sample mean be? **The answer is the p-value.**

In practice: is the claimed value inside or outside our 95% CI (39.6 to 50.4)?

- 39 is outside → our mean falls in the rare 5% → p = .026 < .05 → **reject H_0: the true mean is different from 39.**
- 42 is inside → p ≥ .05 → **cannot reject H_0.** We do not say “the mean is 42” – only “no evidence that it differs from 42”.

Why .05? It is the 5% left outside the middle 95%. A convention, not a law of nature.



Where SPSS Shows It – Explore

MENU PATH

Analyze → Descriptive Statistics → Explore

Variable → Dependent List · Statistics: Descriptives (95% CI by default) · Plots: tick “Normality plots with tests”

In the Descriptives table	What it tells you
Mean	Our best guess for the population mean (the middle)
Std. Error	How much the mean would move in another sample
95% CI for Mean: Lower / Upper Bound	Where the population mean probably lies
Std. Deviation	How spread out the people are (describes the data)
Skewness / Kurtosis	The shape of the distribution (as with FREQUENCIES)

Same menu, same output: the Tests of Normality table (Shapiro-Wilk, Kolmogorov-Smirnov) and the Q-Q plot also come from Explore.

What H_0 Always Looks Like

H_0 ALWAYS…

1 contains "=": one specific value we can centre the bell on (mean = 39, distribution = normal, difference = 0)

2 says "no difference, no effect, no relationship"

3 is assumed true while we calculate p

4 is never proven: we either reject it or fail to reject it

One logic for every test

Today we compared one mean with a claim, and we could see the answer in the CI ourselves.

For harder questions – comparing two groups, or a relationship between two variables – SPSS runs the same process for us. Every test has its own H_0; SPSS calculates p for it.

Our job: know what H_0 says, and read p correctly.

FIRST EXAMPLE, RIGHT NOW: THE NORMALITY TEST

H_0: the distribution does not differ significantly from normal.

Normality Tests

Today we test two variables we know well: age and comminute.

TWO TESTS, ONE QUESTION

H_0: the distribution does not differ significantly from normal

Shapiro-Wilk: usual choice for small samples (n < 50 by convention).

Kolmogorov-Smirnov: for larger samples. SPSS prints both.

READING THE Sig. COLUMN

Sig. is the p-value

Sig. > .05 → no evidence against normality. (Not proof that it is normal.)

Sig. ≤ .05 → the variable departs from normal.

Q-Q plot: points on the line = normal; points bending away = not normal.

Menu: Analyze → Descriptive Statistics → Explore → Plots → tick “Normality plots with tests”

WATCH OUT

Sample size changes the answer

With n ≈ 1,700 the tests flag even tiny departures. Decide with histogram, Q-Q plot, skewness/kurtosis and the test together. p < .05 always means “reject H_0” – whether that is good or bad depends on what H_0 says.

From One Sample to a p-Value

1

Population & sample

One true mean – but unknown. We see one sample; its mean is our best guess.

2

Error

Our mean misses the true mean by some amount. We can never calculate our own error.

3

CLT: the shape

The means of all possible samples form a bell, centred on the true mean.

4

SE: the width

SE = SD ÷ $\sqrt{n}$ – how far a sample mean typically lands from the true mean.

5

95% CI

Our mean ± 1.96 × SE. 95% of sample means fall within 1.96 SE; 5% fall outside.

6

p-value

H_0 claims a value. Outside the CI → $p < .05$ → reject H_0. Inside → cannot reject.

7

Normality test

First use of p in SPSS. H_0: the distribution does not differ from normal.

The same logic is behind every test we will learn:

set up H_0 → calculate p → reject or not.

SPSS does the calculation; we read H_0 and p.